

simon-5501-06-slides

2026-09-29

Topics to be covered

- What you will learn
 - Diagnostic plots
 - R code for diagnostic plots
 - Influence measures
 - R code for influence measures
 - Log transformations
 - R code for log transformations
 - Your homework

The population model

- $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, i = 1, \dots, N$
 - ϵ_i is an unknown random variable
 - Mean 0, standard deviation, σ
 - Often assumed to be normal
 - β_0 and β_1 are unknown parameters
 - b_0 and b_1 are estimates from the sample

Speaker notes

You only have access to data from a sample, but it is important to recognize that the linear regression estimates from the sample are actually estimates of parameters from the population. In the population model, the subscript i goes from 1 to N . We capitalize N to emphasize that it is much larger than lower case n , the size of your sample.

The term β_0 represents the intercept, the population average value of Y when $X=0$. It is not an estimate because the population is fixed (no sampling error). The term β_1 represents the slope, the change in the population average of Y when X increases by one unit. Again, this is a fixed value because the population has no sampling error. The term ϵ represents how much an individual Y value in the population deviates from $\beta_0 + \beta_1$ times the corresponding X value in the population.

If you want to compute confidence intervals and test hypotheses involving β_0 and β_1 , you need to make some assumptions about the ϵ s. You have to assume that the ϵ s are normally distributed, with a mean of zero and a common standard deviation, σ . You also have to assume that the ϵ s are independent of one another.

Violations of this model

- Nonlinearity
- Heterogeneity
- Non-normality
- Lack of independence

Speaker notes

There are four things that you need to check. The first is non-linearity. Maybe the relationship between X and Y is more complex than a straight line.

The second is heterogeneity. Each epsilon has to have the same standard deviation.

The third is non-normality. The distribution of the epsilons might not follow a bell shaped curve.

The fourth is lack of independence. The epsilons might be related to one another.

Using residuals for diagnostic plots

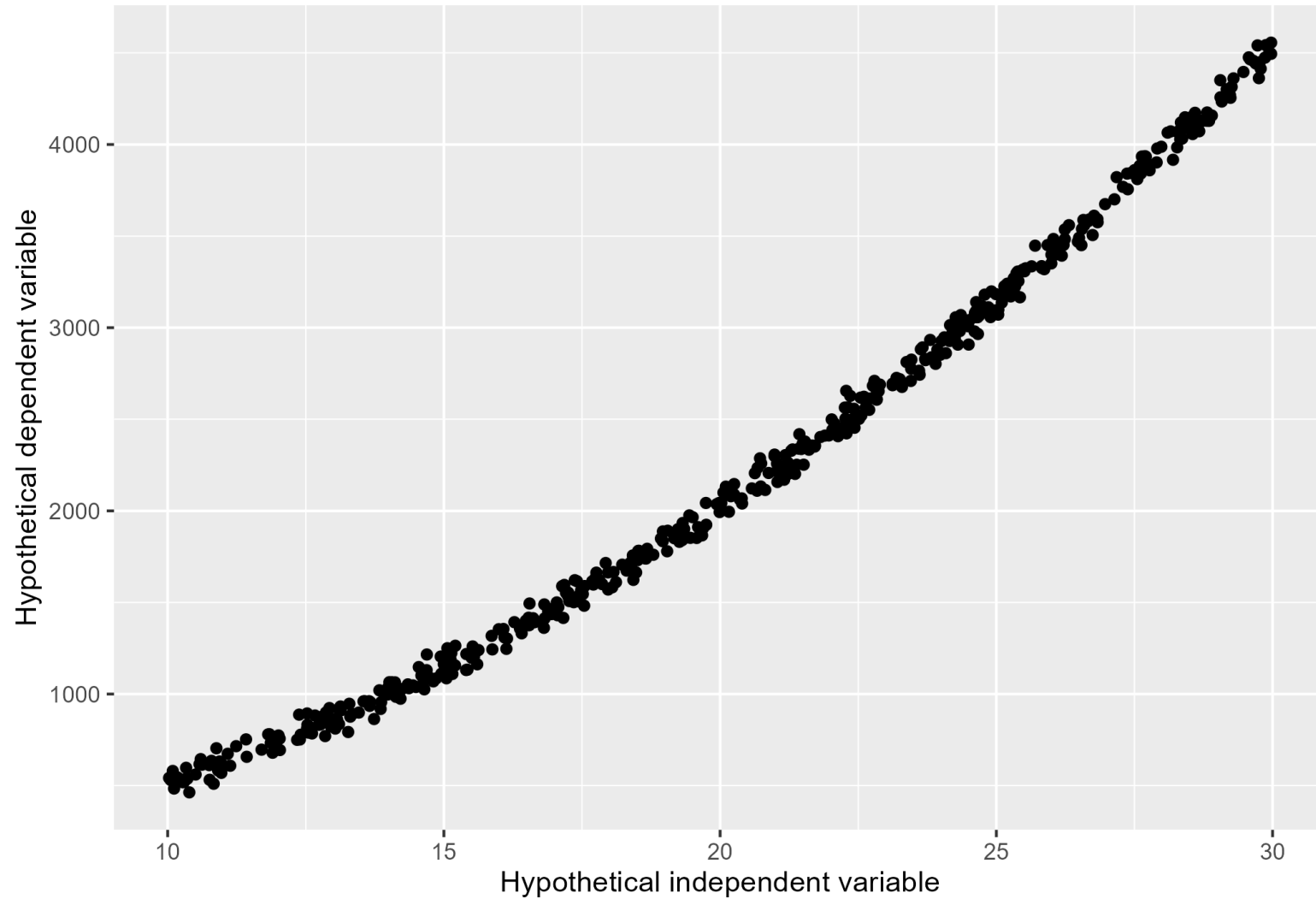
- ϵ_i is unknown, but e_i is known
 - $\epsilon_i = Y_i - (\beta_0 + \beta_1 X_i)$
 - $e_i = Y_i - (b_0 + b_1 X_i)$
 - Are there problems with the e_i
 - Indirect evidence of problems with the ϵ_i
- Residuals show patterns more clearly than the original data

Speaker notes

You can not get a direct assessment of whether the epsilons have problems with non-linearity, heterogeneity, non-normality, or lack of independence. The epsilons are unknown, unless you have access to the entire population. You can compute the residuals from the sample using the estimated regression coefficients. It's not perfect, but if you notice problems in how the residuals behave, that is indirect evidence of problems with the epsilons.

A key point that you will see if that patterns in the dependent variable are often subtle and easy to overlook. This can often occur when there is a strong linear trend. The patterns that indicate problems with the assumptions are often far more apparent when you examine the residuals.

Diagnosing non-linearity, 1

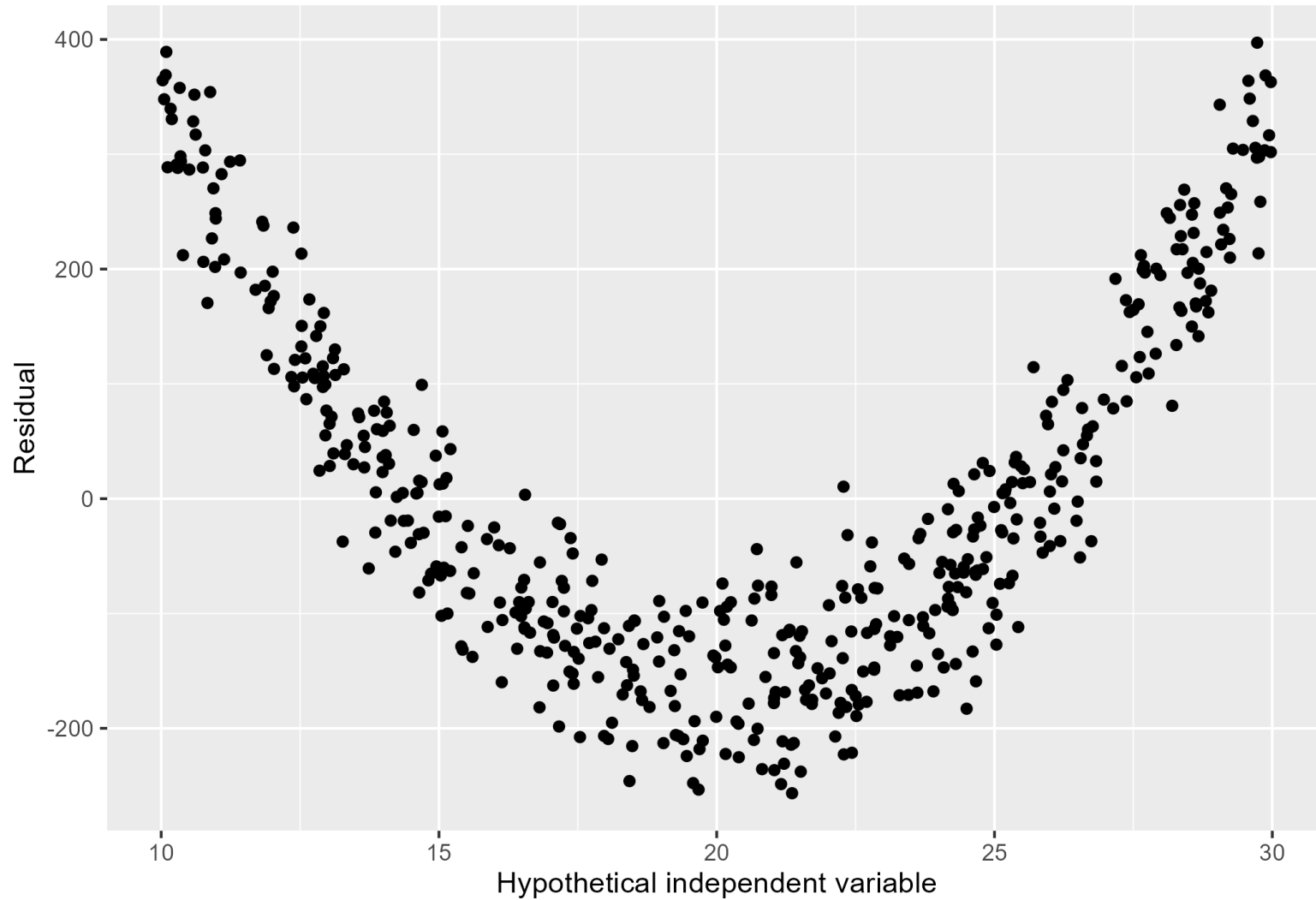


Simon, 2024-09-23

Speaker notes

Here is some artificial data. It shows a strong relationship, and if you look closely, you might detect a slight bend in the data.

Diagnosing non-linearity, 2



Simon, 2024-09-23

Speaker notes

Here is the same data with the residuals on the Y-axis. A slight and very subtle bend becomes hard-to-miss evidence of non-linearity.

Remedies for non-linearity

- Largely beyond the scope of this class
 - Ignore the non-linearity
 - Add a quadratic term
 - Consider a non-linear model
 - Use a spline model

Speaker notes

If you see problems with non-linearity, you have several options. Some of these are beyond the scope of this class.

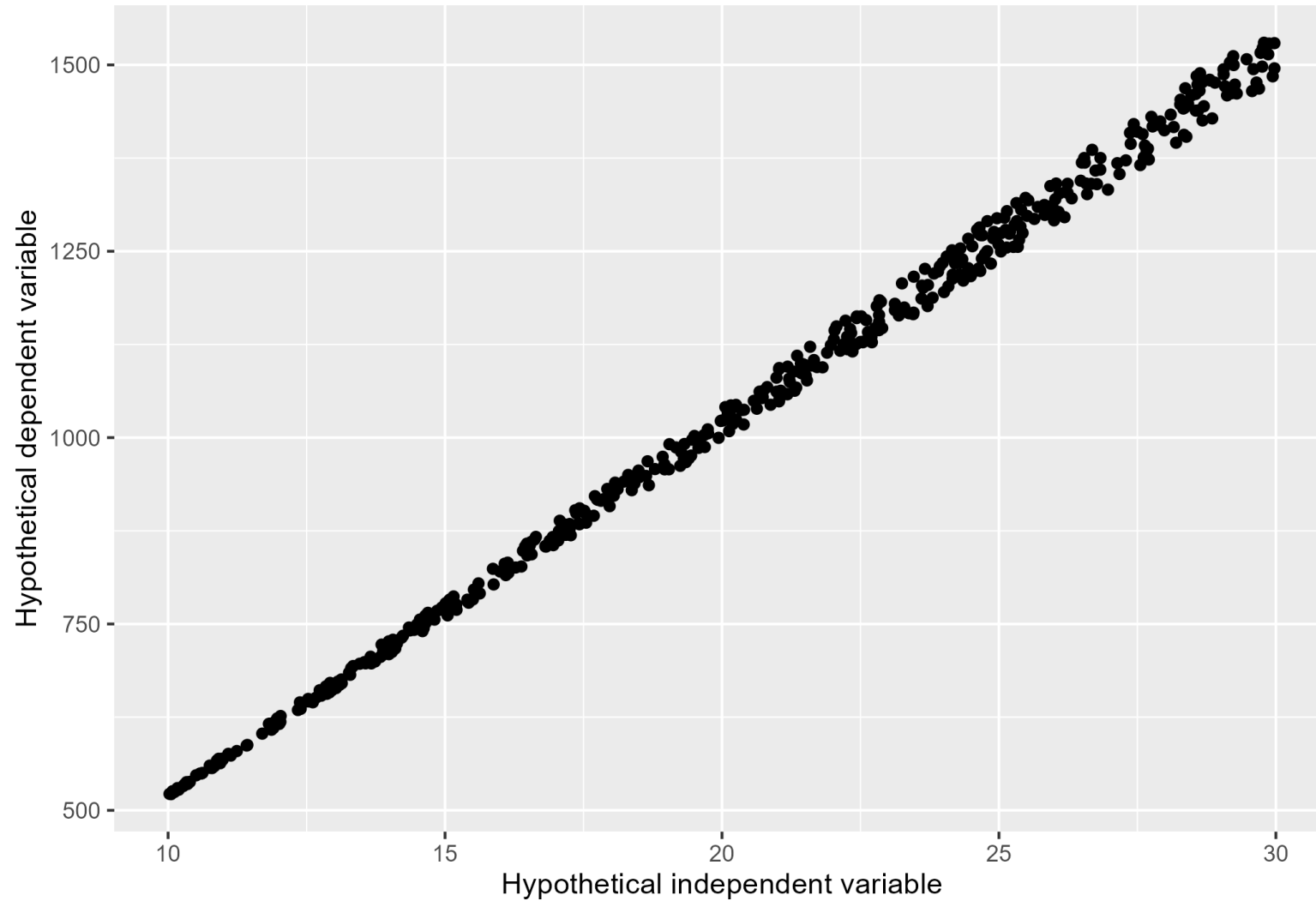
First, if your linear trend is very strong, maybe you can ignore the non-linearity. It means a less than perfect fit, but maybe it would be close enough in a real-world setting.

Second, you might find that a linear term plus a quadratic term does a good job. I will show how to add a quadratic term in the next module.

Third, you might have some information based on your knowledge of research context that would allow you to specify a non-linear function. This is a topic that I might cover in MEDB 5502, Applied Biostatistics II.

Fourth, you might consider a spline model. This is also a topic that I might cover in MEDB 5502.

Diagnosing heterogeneity, 1

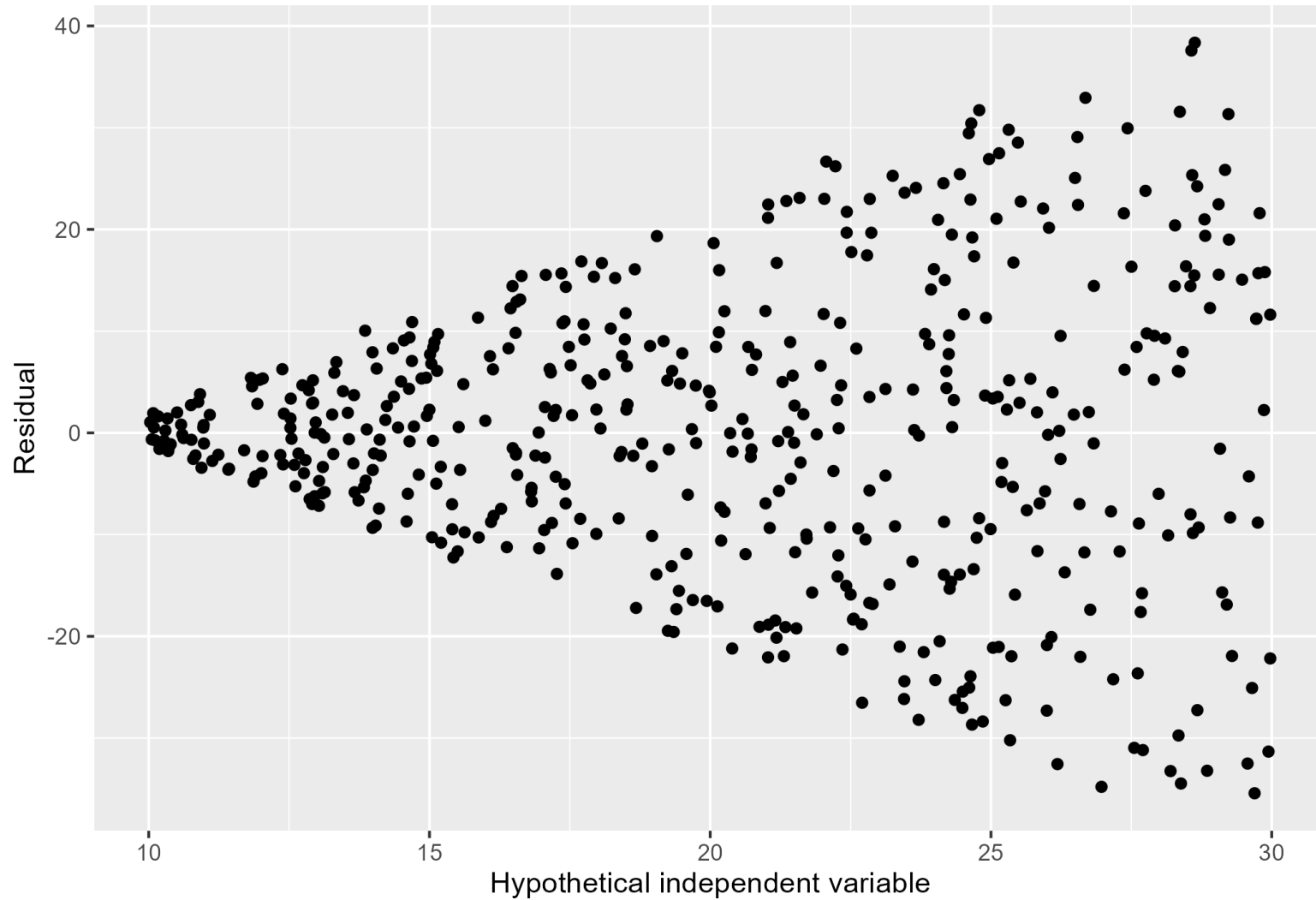


Simon, 2024-09-23

Speaker notes

Here's another example of some artificial data. You can see a bit more variation on the high end of the X axis.

Diagnosing heterogeneity, 2



Simon, 2024-09-23

Speaker notes

The change in variation is much more apparent when you look at the residuals. This fanning out pattern is quite common in residual diagnostics.

Remedies for heterogeneity

- Ignore the heterogeneity
- Log transformation
 - Especially useful if larger values have more variation
- Weighted regression

Speaker notes

There are several choices facing you if you observe heterogeneity in your data.

First, a small amount of heterogeneity can be safely ignored. If the variation differs by a factor of two or less, then it may not be worth addressing.

Second, the log transformation can sometimes help. It stretches the small values and squeezes the large values. This can help quite a bit when larger values in your data exhibit larger amounts of variation. I'll show how the log transformation works later in this lecture.

Third, you can assign weights giving greater emphasis to data points with more precision (smaller variation) and lesser emphasis to data points with less precision (larger variation). This is not done too often in practice, but it can sometimes greatly improve the overall stability of your estimates.

Diagnosing non-normality, 1

- Independent variable (X), non-normality is okay
- Dependent variable (Y), non-normality is okay
- Population residual (ϵ), non-normality is not okay
 - Confidence intervals, hypothesis tests not valid
 - Less of a concern with large sample sizes

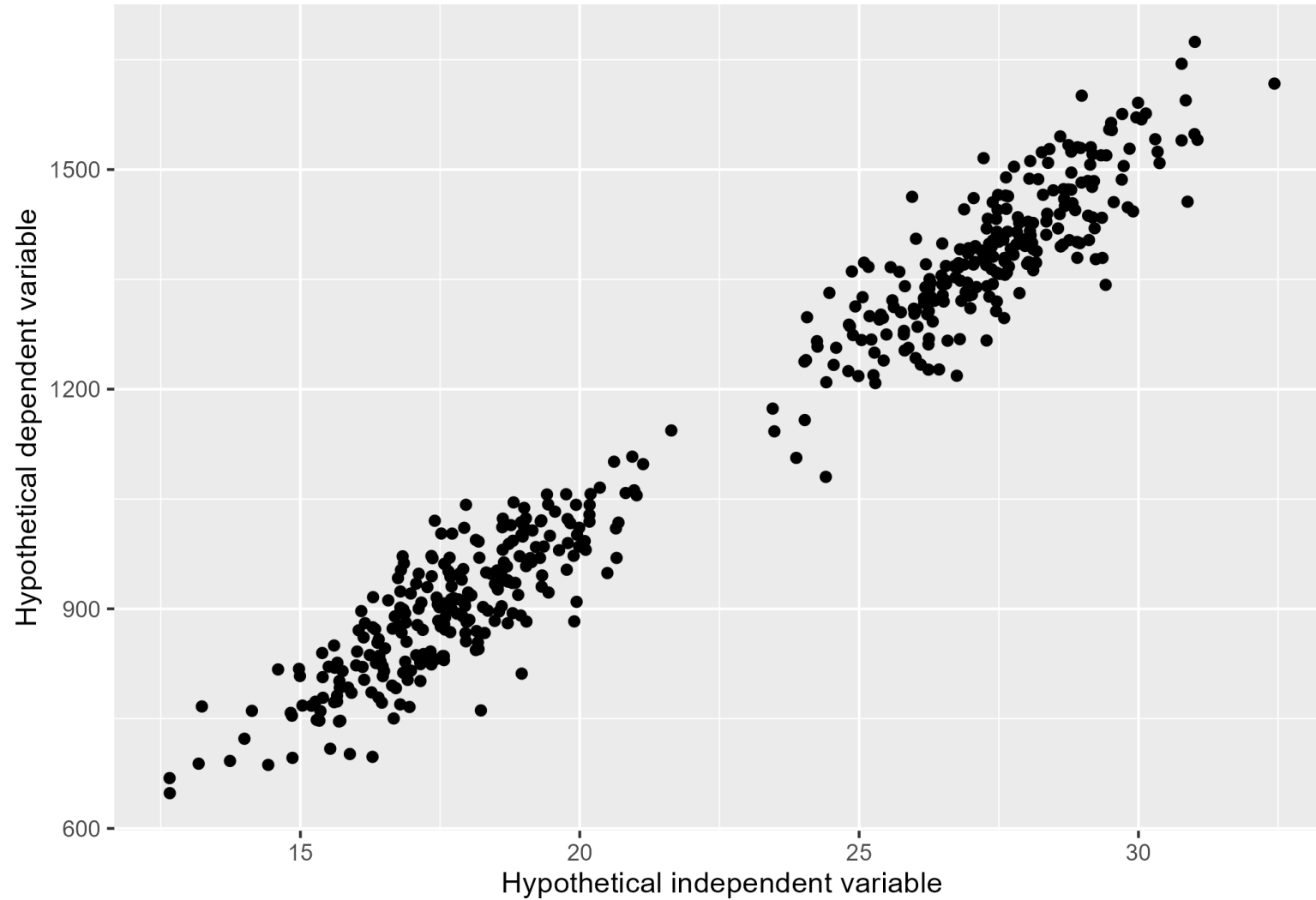
Speaker notes

When you think about non-normality, you must be careful. A non-normal independent variable is okay. It could be bimodal, skewed, or have outliers. None of this matters.

A non-normal dependent variable is also okay. This occurs often when the independent variable has some strange non-normal pattern.

The only thing that needs to be normal is your epsilons. And you diagnose that assumption of normality with the residuals.

Diagnosing non-normality, 2

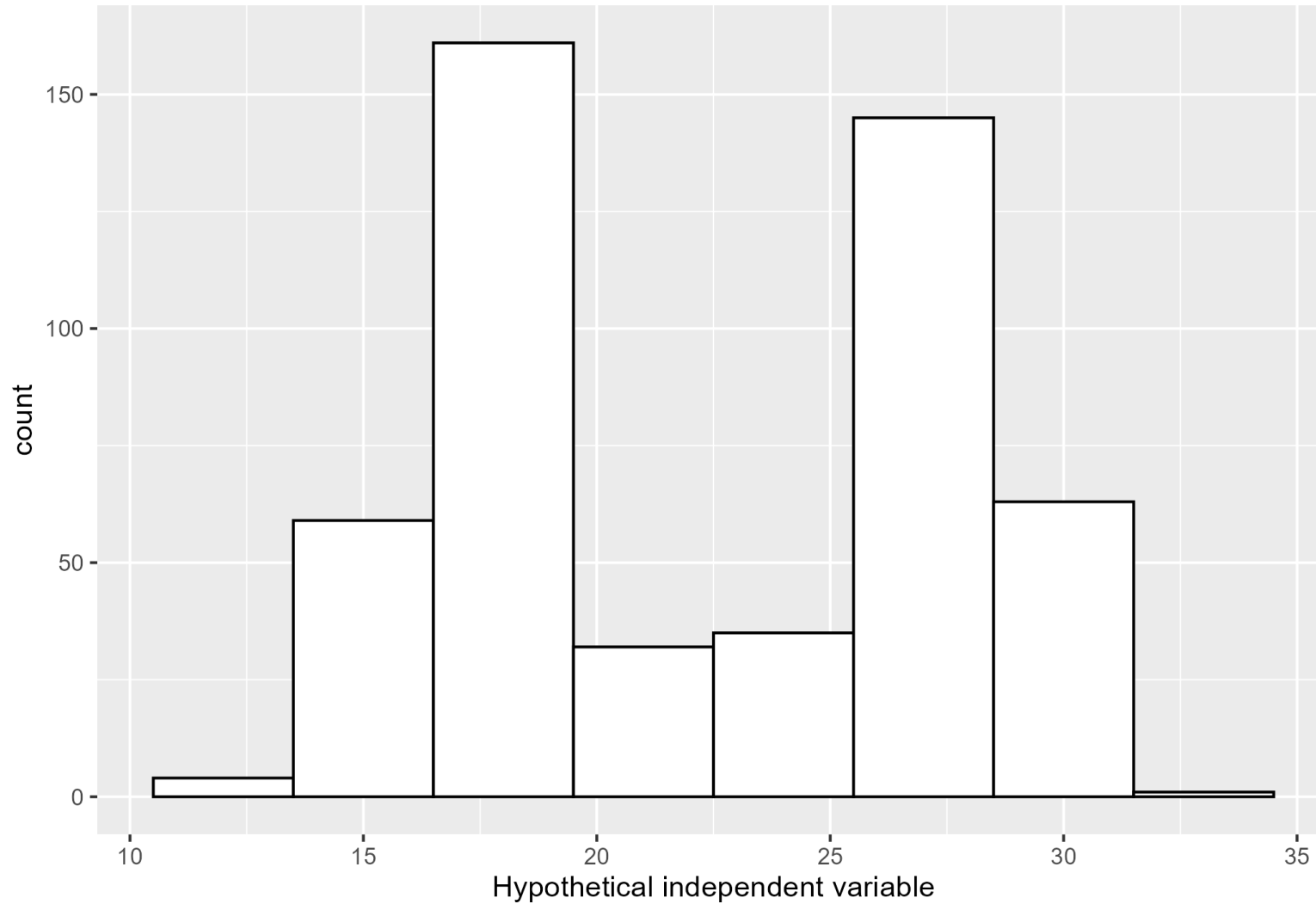


Simon, 2024-09-23

Speaker notes

Here is some more artificial data to illustrate where normality is an issue. Notice a gap near the middle. This can happen for a variety of reasons.

Diagnosing non-normality, 2

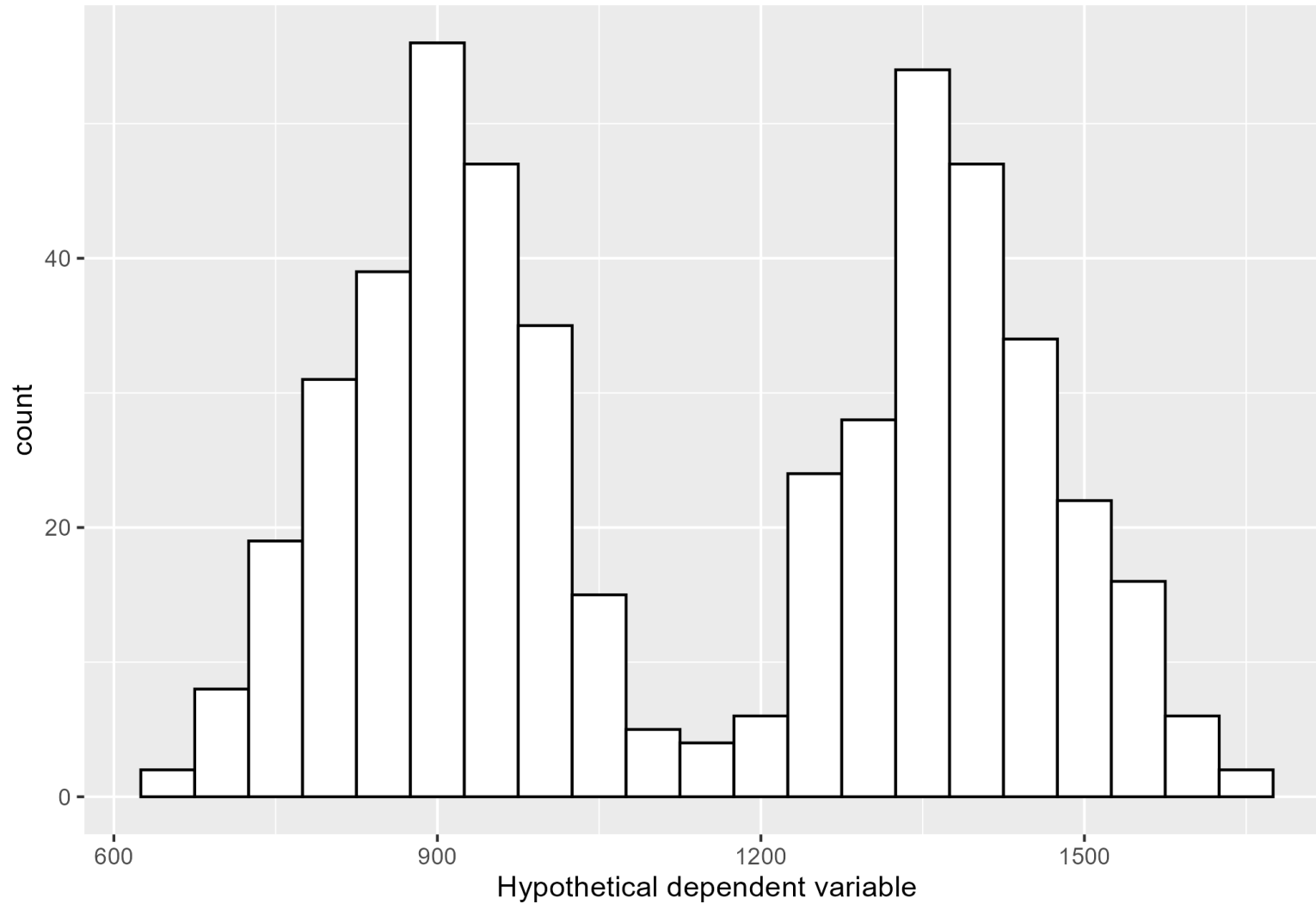


Simon, 2024-09-23

Speaker notes

The histogram of the independent variable shows a strong bimodal pattern. This is not a violation of assumptions.

Diagnosing non-normality, 3

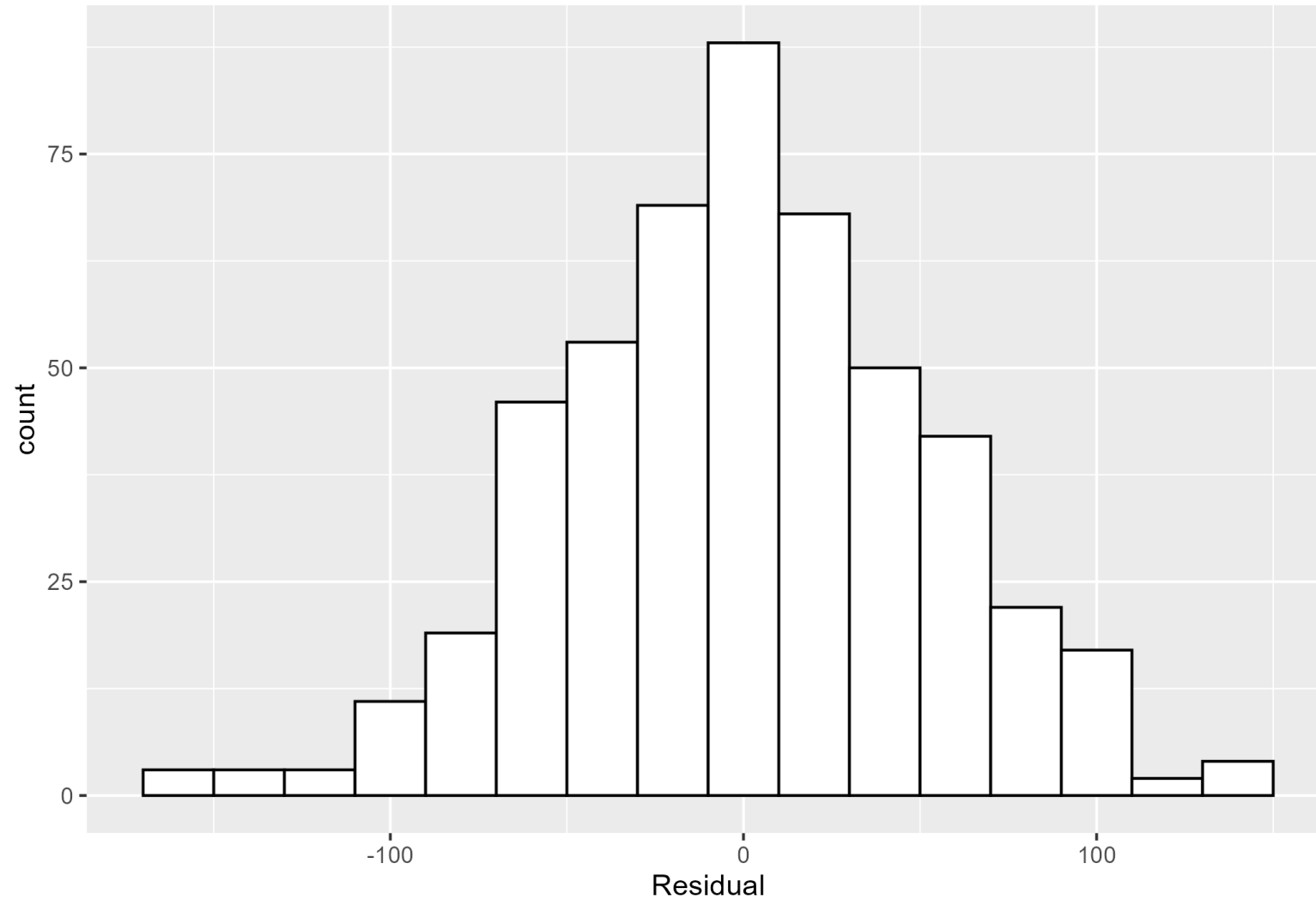


Simon, 2024-09-23

Speaker notes

The histogram of the dependent variable also shows a strong bimodal pattern. This is not a violation of assumptions.

Diagnosing non-normality, 4



Simon, 2024-09-23

Speaker notes

It is only a plot of the residuals that is important. In this example, the residual histogram is fairly close to a bell shaped curve, indicating that non-normality is not a concern.

Remedies for non-normality

- Ignore the non-normality
 - Large sample sizes
 - Certain types of non-normality
- Log transformation
 - Especially useful for right-skewed data
- Generalized linear models

Speaker notes

If you see non-normality in the residuals, you have several options.

First, you can sometimes ignore the non-normality. Non-normality is less of a concern for large sample sizes because of the Central Limit Theorem. How large is large? it depends on the type of non-normality. With certain types of non-normality, especially light tailed distributions, you're fine with a few dozen data points. For highly skewed data, or data with many outliers, you may need hundreds of data points before you can safely ignore non-normality.

Second, the log transformation may help here. It works best with fixing problems associated with right-skewed data. Recall that right skewed data has a greater tendency to produce outliers on the high end. This can be fixed by the tendency of the log function to square large values and stretch small values.

Lack of independence

- Often judged qualitatively
- Clustering
 - Litter effects
 - Family members
 - Clinic effects
 - Multi-center trials
- Infectious diseases
- Time sequence effects

General thoughts on assumption tests

- Don't over-interpret small departures from assumptions
- Normality is not an important assumption with large sample sizes
- A violation of assumptions is actually an opportunity
 - A more complex model could greatly improve things
- Linear regression is always okay as a descriptive tool

Speaker notes

In general, you should take remedial action only for large departures from the underlying assumptions. The main reason for this is that sampling error might produce a false warning.

In particular, the normality assumption is not really as important as the others. Do worry about highly skewed data and extreme outliers. Other types of non-normality are less concerning. Even large deviations from normality are not a serious concern for large sample sizes (several hundred might be sufficient).

Don't think of a violation of assumptions as a death warrant or an insurmountable barrier. Think of it as an opportunity. The relationship between the independent variable and the dependent variable needs a more complex model. This is more work but it is also an opportunity to produce interesting and valid results.

Break #1

- What you have learned
 - Diagnostic plots
- What's coming next
 - R code for diagnostic plots

Live demo, part 1

- This section uses - a program, [simon-5501-06-demo.qmd](#) - a data set, [albuquerque-housing.csv](#) - with a data dictionary, [albuquerque-housing.yaml](#)

Break #2

- What you have learned
 - R code for diagnostic plots
- What's coming next
 - Influence measures

Influence measures

- Does one data point dominate the regression fit?
 - Extreme in the X direction
 - Deviation from the trend line
- Influential data points are NOT a violation of assumptions

Speaker notes

Linear regression has been around for hundreds of years. Recently (meaning 1980s), reserachers have been paying attention to linear regression models where one data point dominates the regression fit.

You can think of these influential data points as outliers, just like Elon Mush was an outlier when looking at average incomes. But you can be an outlier in two different ways. You can be extreme in the X direction, or you can deviate from the trend line in the y direction.

Influential data points are not a violation of assumptions. If you find an influential data point, study it carefully and make sure that it is not a typo or an ineligible patient who sliped by the initial screen.

Studentized deleted residual

- $\hat{Y}_{(i)}$ is the predicted value of Y at $X = X_i$ using all the data except (X_i, Y_i)
- $e_{(i)} = \hat{Y}_{(i)} - Y_i$ is the deleted residual
- $se(e_{(i)})$ is the standard error
 - or, the estimated standard deviation of the deleted residual
- $e_{(i)} / se(e_{(i)})$ is the studentized deleted residual
 - Investigate if $> +3$ or < -3

Speaker notes

One way to measure whether a data point is influential is to see what happens to the predicted value when that data point is removed. You can do this for every data point in your sample. It seems like that would take a lot of work, but computers these days are very fast, so even having thousand of data points will not slow these calculations down too much.

The deleted residual is a residual computed without the potentially influential data point. You can compute the standard error of the deleted residual. In general, the standard error is the estimated standard deviation of a statistic. When you divide a statistic by its standard error, you get something that is small if it is between plus or minus three.

Leverage

- $h_{ii} = \frac{1}{n} + \frac{(X_i - \bar{X})^2}{\sum (X_j - \bar{X})^2}$
 - Investigate if $> 3 \cdot 2/n$

Speaker notes

Leverage is a measure of deviation in the X direction, or outliers with the independent variable. The average leverage is $2/n$. A large value (> 3 times the average value of $3 \cdot 2/n$) is worth investigating.

Cook's distance

- $D_i = \frac{\sum (\hat{Y}_j - \hat{Y}_{j(i)})^2}{(k+1)\sqrt{MSE}}$
 - where k is the number of independent variables
 - investigate if > 1

Speaker notes

Cook's distance is a measure of how much the predicted value changes when a point is removed. it is measured relative to a constant times the square root of MSE. Think of the denominator as a measure of noise. If the change stands out relative to noise, if it is greater than 1, then that data point is influential. Cook's distance is a measure that combines leverage with the studentized deleted residual. Cook's distance is largest if both the leverage is large and the studentized deleted residual is not close to zero.

Break #3

- What you have learned
 - Influence measures
- What's coming next
 - R code for influence measures

Live demo, part 2

- This section is a continuation of the demo. It uses the same files.
 - `simon-5501-06-demo.qmd`
 - `albuquerque-housing.csv`
 - `albuquerque-housing.yaml`

Break #4

- What you have learned
 - R code for influence measures
- What's coming next
 - Log transformations

The log transformation

- Useful for correcting non-normality
 - Only if skewed right
- Useful for correcting heterogeneity
 - Only when larger values have larger variation

Which logarithm?

- Natural logarithm, $\ln$, or $\log_e$
 - R function is just `log()`
 - Difficult to interpret
 - Simple from a mathematics perspective
- Base 10 logarithm, $\log_{10}$
 - R function is `log10()`
 - Easiest to interpret, powers of 10
- Base 2 logarithm, $\log_2$ - R function is `log2()` - Powers of 2, common in genomics

Back transformation, 1

- Puts regression coefficients back on original scale
- Antilog inverts log
 - $\exp()$ inverts $\log()$
 - $10^{\wedge}$ inverts $\log_{10}()$
 - $2^{\wedge}$ inverts $\log_2()$

Back transformation, 2

- $Y_i^* = \log_{10}(Y_i)$ or $\log_2 Y_i$ or $\ln(Y_i)$
- $\hat{Y}_i^* = b_0^* + b_1^* X_i$
- $b_0 = \text{antilog}(b_0^*)$
 - Estimated average value of Y when X=0
 - Actually a geometric mean

Back transformation, 3

- $b_1 = \text{antilog}(b_1^*)$
 - Estimated average relative change in Y when X increases by one unit
 - 1.1 represents a 10% increase
 - 0.9 represents a 10% decrease

Further details

- pmean website
 - [Log transformation, 2002-10-11][ref-simon.2002]

Break #5

- What you have learned
 - Log transformations
- What's coming next
 - R code for log transformations

Live demo, part 3

- This section is another continuation of the demo. It uses the same files.:
 - `simon-5501-06-demo.qmd`
 - `albuquerque-housing.csv`
 - `albuquerque-housing.yaml`

Break #6

- What you have learned
 - R code for log transformations
- What's coming next
 - Your homework

Your homework

- [simon-5501-06-directions.md](#)

Summary

- What you have learned
 - Diagnostic plots
 - R code for diagnostic plots
 - Influence measures
 - R code for influence measures
 - Log transformations
 - R code for log transformations
 - Your homework